

Early Childhood Development and the Next 1000 Days 2



The cost of not investing in the next 1000 days: implications for policy and practice

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Building on the evidence from the first paper in this Series highlighting the fundamental importance of healthy and nurturing environments for children's growth and development in the next 1000 days (ages 2–5 years), this paper summarises the benefits and costs of key strategies to support children's development in this age range. The next 1000 days build on the family-based and health-sector based interventions provided in the first 1000 days and require broader multisectoral programming. Interventions that have been shown to be particularly effective in this age range are the provision of early childhood care and education (ECCE), parenting interventions, and cash transfers. We show that a minimum package of 1 year of ECCE for all children would cost on average less than 0·15% of low-income and middle-income countries' current gross domestic product. The societal cost of not implementing this package at a national and global level (ie, the cost of inaction) is large, with an estimated forgone benefit of 8–19 times the cost of investing in ECCE. We discuss implications of the overall evidence presented in this Series for policy and practice, highlighting the potential of ECCE programming in the next 1000 days as an intervention itself, as well as a platform to deliver developmental screening, growth monitoring, and additional locally required interventions. Providing nurturing care during this period is crucial for maintaining and further boosting children's progress in the first 1000 days, and to allow children to reach optimal developmental trajectories from a socioecological life-course perspective.

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This is the second in a Series of two papers on early childhood development and the next 1000 days

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Introduction

The past decade has seen remarkable growth in global, regional, and national efforts to upscale early childhood development programmes,¹ providing unprecedented opportunities to reduce early life disparities and improve children's health and wellbeing in both the short and long term.² Major milestones in the global efforts towards improving children's early life experiences and early life outcomes include the adoption of the Sustainable Development Goals (SDGs) in 2015, the launch of both the global Early Childhood Development Action Network in 2016 and the Nurturing Care Framework (NCF) in 2018, the G20's 2018 commitment to investments in the early years,³ as well as the launch of regional initiatives, such as the Africa Early Childhood Network, the Asia-Pacific Regional Network for Early Childhood, and the Arab Network for Early Childhood.

This Series paper builds on evidence from the first paper⁴ in this Series highlighting the fundamental importance of nurturing care for healthy growth and development in the next 1000 days. Through its main goal of identifying the most effective strategies to improve early childhood development in this age range, this Series paper builds on previous work of the 2007, 2011, and 2017 *Lancet* Series that considered the first 5 years of children's lives with a particular focus on the first 3 years^{5–8} as well as research assessing interventions in the first and next 1000 days.^{9–11}

This Series paper builds on the mapping review of interventions presented in the first paper, and focuses on the evidence regarding the documented effectiveness of key interventions in the next 1000 days. We also present

Key messages

- The next 1000 days are a key period for children's development, and providing adequate support for children in this period is of crucial importance for their long-term development
- Robust and rapidly growing evidence highlights the short-term and long-term benefits of interventions specific to the next 1000 days
- High-quality ECCE programmes offered in this period can improve short-term and long-term cognitive and academic outcomes
- Parental educational programmes have been shown to yield sizeable improvements in children's developmental outcomes
- Although cash-transfer and nutritional interventions yield more moderate improvements compared to ECCE, they contribute positively to childhood outcomes, indicating their potential as complementary or synergistic strategies
- ECCE services can serve as key platforms for delivering essential complementary programmes and services such as the promotion of healthy behaviours, growth and developmental monitoring, and nutritional supplementation
- Despite the proven benefits, access to ECCE remains low globally, particularly in low-income countries
- The costs of providing ECCE and other health and nutritional services vary substantially across countries, but are on average low relative to national incomes, with an estimated total cost of less than 1% of GDP for one year of universal ECCE in LMICs
- The potential benefits of providing at least one year of ECCE to all children are on average 8–19 times larger than the cost of implementing these programmes across LMICs
- Investment in children in this age group can help countries achieve the ambitious SDG targets for health, education, and equity, complementing investments made in the first 1000 days

ECCE=early childhood care and education. GDP=gross domestic product. LMICs=low-income and middle-income countries. SDG=Sustainable Development Goals.

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the latest estimates of the relationship between enrolment in early childhood care and education (ECCE) programmes and long-term outcomes using data from intervention studies and from nationally representative observational data. We go on to assess the costs of key interventions for this age group and estimate the global

cost of inaction (ie, the societal loss in terms of unrealised gains faced by countries if a minimal set of key interventions is not expanded to cover all children). Recognising that ECCE coverage is far from universal today despite the strong evidence supporting its effectiveness, we focus on the universal provision of at least 1 year of ECCE as the most obvious and immediate target in our cost of inaction analysis. As outlined in the first paper in this Series, ECCE can be a key platform for covering multiple domains of nurturing care in the next 1000 days. Indeed, as the first paper already highlighted, and further underlined in the discussion at the end of this paper, a range of complementary, developmental, and context-specific home-based and community-based interventions will be needed in practice to ensure nurturing care along all dimensions in this age range.

What works for promoting healthy child development in the next 1000 days?

Effectiveness studies targeting the next 1000 days have primarily focused on four additional types of interventions: cash transfers, ECCE, nutritional interventions, and multicomponent interventions.⁹ Through a comprehensive literature review, we were able to identify 100 studies assessing the effects of these interventions in the next 1000 days on a range of child outcomes, yielding a total of 951 impact estimates across all global regions and income levels. Here we provide an overview of the main impact patterns. Further details on inclusion criteria, methods, and studies identified are provided in the appendix (pp 1–12).

Cash transfers to caregivers or families are generally set up with the primary objectives of reducing poverty among disadvantaged populations and of creating more equitable early conditions from a socioecological life-course perspective. Rather than targeting the child directly, cash transfers aim to support children's development by improving the overall wellbeing of families.⁷ In some countries, transfers are made unconditionally to eligible households; in others, transfers are made conditionally on specific behaviours presumed to be beneficial for children, such as attendance in ECCE programmes or completion of

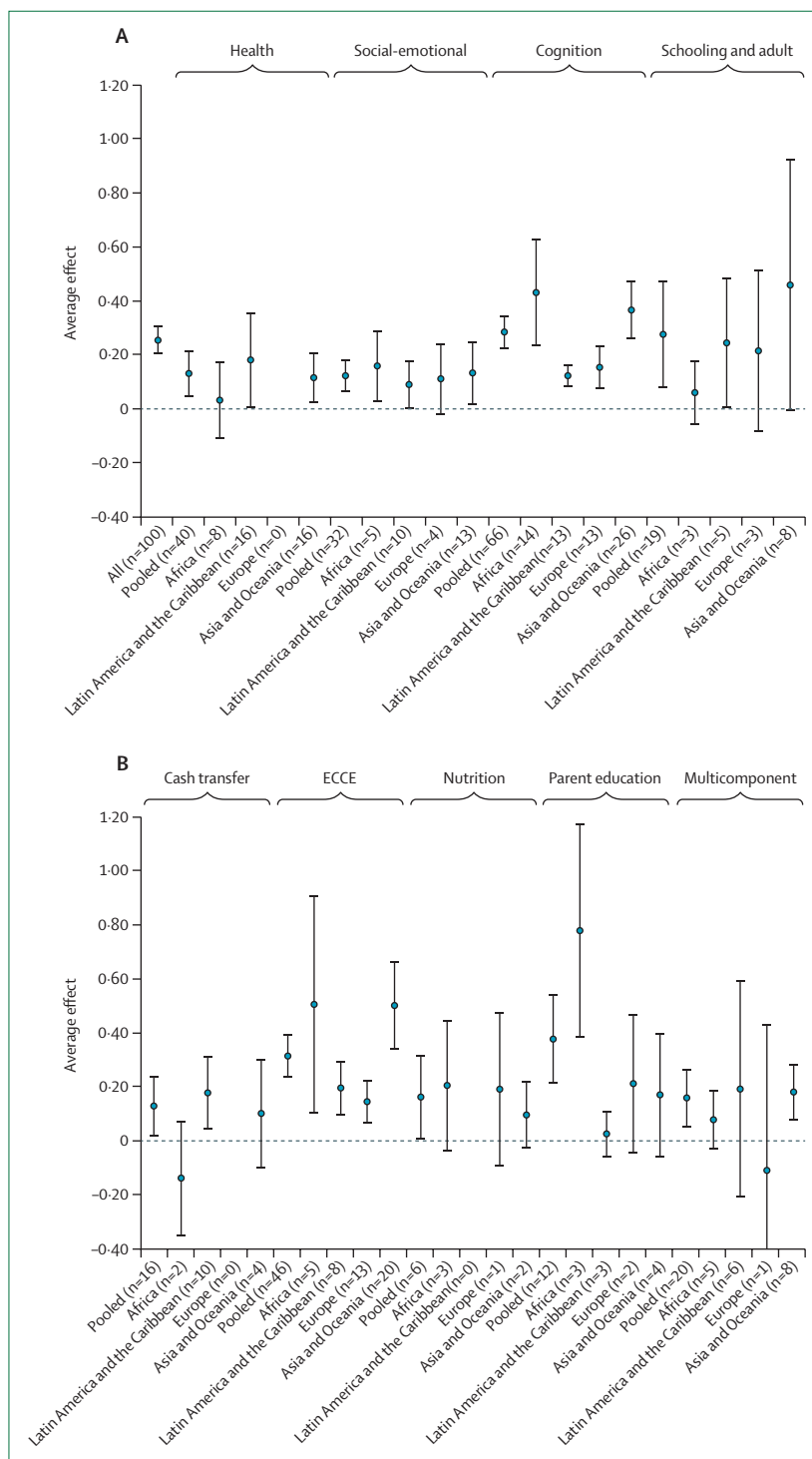


Figure 1: Average effect of interventions in the next 1000 days aggregated by intervention type, outcome, and region

Datapoints are mean effect estimates with error bars showing 95% CIs. Results are based on 100 studies including children aged 3–6 years, conducted outside of the USA and published by July 30, 2020. Mean effect sizes are reported averaged by region, type of intervention, and outcomes in standard deviation units, and were calculated using weight-adjusted random effects meta-analyses using the methods described by McEwan and colleagues.^{13,14} Random effects variance components were calculated using a restricted maximum likelihood estimator.¹⁵ Panel A shows mean effects by outcome and region, pooling all interventions. Panel B summarises mean effects by region and type of intervention, pooling across all outcomes. Further details on studies and methods can be found in the appendix (pp 1–14). ECCE=early childhood care and education.

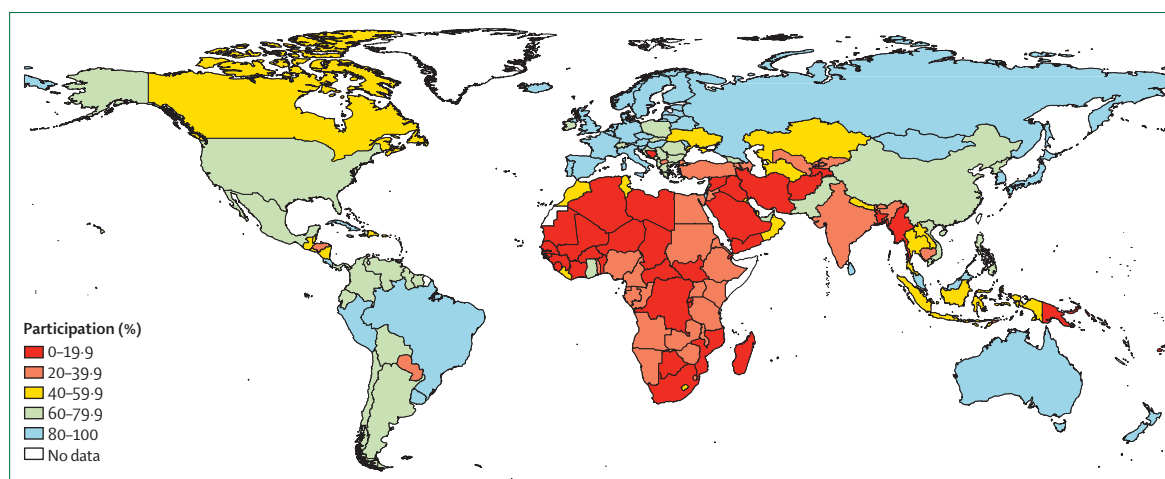


Figure 2: Current global ECCE participation rates for children aged 3–5 years

ECCE participation was measured using net enrolment in ECCE or attendance to ECCE settings when data on net enrolment were not available. Overall, both definitions of ECCE participation consider a wide set of programme types, including public, private, and non-profit programmes. See McCoy and colleagues¹⁸ for further details. ECCE=early childhood care and education.

routine child-health check-ups.¹² As shown in figure 1, cash transfers moderately improved child developmental and health outcomes by 0.13 SD (mean estimated effect size [ES]=0.13, 95% CI 0.02 to 0.24), with similar effects on cognition (0.13, 0.00 to 0.25), somewhat larger average effects on social-emotional development (0.23, -0.04 to 0.50), and smaller effects on health (0.12, -0.03 to 0.26; appendix pp 5–7). To situate these effects in context, mean Programme for International Student Assessment (PISA) scores for high-income countries (HICs) are 0.8 SD higher than those of low-income countries;¹⁶ therefore, an effect size of 0.13 SD would be equivalent to 16% of the gap between HICs and low-income and middle-income countries (LMICs). Cash-transfer programmes have been evaluated in diverse settings such as Brazil, Mexico, Nepal, and Ethiopia. At the regional level, effects appear largest in Latin America and the Caribbean (0.18, 0.04 to 0.31), where these programmes have been most widely used. Conditional programmes show on average slightly larger effects than unconditional ones (0.15, -0.03 to 0.33; appendix p 7). Common conditions attached to transfers in Latin America and the Caribbean are completion of recommended vaccinations and school attendance (eg, Bolsa Escola in Brazil), use of preventive health visits (eg, Familias en Acción in Colombia), or growth monitoring (eg, Juntos programme in Peru). Conditional programmes have been shown to improve not only parental health knowledge and use of preventive health services, but also child nutrition and child health more generally.¹³ Unconditional cash transfers generally allow households to expand their expenditure on all essential items, including food, supplies, and care,¹⁷ all of which are beneficial for children between 2 years and 5 years.

ECCE programmes include home-based, community-based, and centre-based programmes that provide care, educational supports, or a combination to children. ECCE programmes vary by age, and now exist in virtually all countries. Provided through a mix of private organisations, non-profit organisations, and public sector agencies, ECCE programmes are non-compulsory in most settings and ECCE participation rates vary widely across country-income groups. 54% of pre-primary school children aged 3–5 years attend ECCE globally; at least 80% of children attend in most HICs compared with less than 20% in most low-income countries, and there are often large socioeconomic disparities in the access to, and quality of, ECCE programmes (figure 2).^{19–22}

Various reviews^{9,23,24} as well as single-country and multi-country analyses^{25–30} have assessed the effectiveness of national ECCE programmes, such as those in Argentina, Chile, Denmark, Germany, Peru, and Uruguay, and comprehensive ECCE programmes integrating health monitoring and nutrition, such as those in Colombia, Mauritius, and the Philippines. Substantial evidence is also available for small-scale ECCE interventions, such as those in Bangladesh, Botswana, Cambodia, and Mauritius (appendix pp 9–12). Consistent with a recent meta-analysis conducted by the World Bank,³¹ we find that ECCE programmes have on average large positive effects on children (0.31, 0.23–0.39; figure 1). These effect sizes are equivalent to about 40% of the difference in PISA outcomes between low-income countries and HICs. Effects are particularly large for schooling and for outcomes such as grade repetition, dropping out of school, and educational attainment (0.40, 0.11–0.70; appendix pp 5–7). Smaller but still positive average effects were found on the health (ES=0.14, 0.04–0.24) and social-emotional domains (ES=0.13, 0.04–0.22; figure 1). Larger average effects were found for

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See Online for appendix

programmes implemented by non-governmental organisations relative to government-run programmes, after accounting for region and income level (appendix p 9). These differences are possibly related to difficulties with delivering quality interventions at scale^{32,33} and reflect a need to monitor and sustain quality (both structural and process quality, including learning materials and evidenced-based and play-based pedagogy, among other aspects)^{34–39} across ECCE expansion efforts.³⁴

The effects of ECCE programmes presented here show remarkable consistency with evidence from the USA, which dominates the research landscape in this age range. Meta-analyses of USA-based studies show improvements between 0·1 and 0·4 SDs in children's development and learning outcomes due to participation in early educational programmes.^{40–44} Although positive effects of ECCE programmes generally weaken as children enter primary school in the USA,⁴² with a few even reversing in direction,⁴⁵ evaluations of USA-based model interventions such as the Perry Preschool, Abecedarian, and Chicago Child-Parent Center programmes,^{46–48} as well as more modern ECCE interventions such as Head Start,⁴⁹ and Boston's, Oklahoma's, and New Jersey's universal preschool programmes,^{50–52} have found substantial and persistent positive long-term effects across various developmental domains. Many of these programmes also feature high-quality components, together with integrated parenting and nutritional components. Documented improvements in educational attainment, adult earnings, health, and crime^{43,53,54} suggest financial returns of ECCE programmes between US\$2·5 and \$16 for every initial dollar of investment in the USA, with particularly high returns among the most disadvantaged populations,^{47,55} as well as additional multigenerational benefits for the offspring of programme beneficiaries.⁵⁶ Positive effects appear most likely to persist when ECCE programmes target more than just basic numeracy and reading skills (eg, foundational competencies such as problem solving), as well as when there is alignment between the high-quality environments that children experience across ECCE programmes, primary schooling, and secondary schooling.^{57–61}

Parenting interventions have been identified as key strategies for the first 1000 days,⁸ and generally are implemented up to 3 years in many settings. Figure 1 also reports very positive effects for parenting programmes that include this age group. Many of these studies are more recent and not necessarily standalone programmes. These include programmes in Armenia (communication), Australia (with fathers), Brazil (home visiting), Hong Kong (home visiting), and Kenya (book sharing), among others. Consistent with the positive effects of these programmes observed during the first 1000 days, parenting programmes in the next 1000 days are shown to improve child outcomes by 0·38 SD (0·21 to 0·54; figure 1), with particularly strong effects observed on cognitive outcomes (0·40 SD, 0·21 to 0·60;

appendix p 6) and positive behavioural outcomes (0·15 SD, –0·01 to 0·31; appendix p 6) in children.

Nutritional programmes during the next 1000 days are common in many LMICs to address continued exposure to hunger, malnutrition, and growth faltering.⁶² Stunting rates are generally higher in the next 1000 days than in the first 1000 days (appendix p 14),⁶³ highlighting the need for screening and interventions, especially as new evidence suggests that catch-up growth in the next 1000 days is possible.^{64–66} The first paper in this Series highlights the potential positive effects of combined nutritional interventions and school feeding programmes, particularly for undernourished children.^{67–69} We identified nutritional studies from Ethiopia (food aid),⁷⁰ Indonesia (dietary supplement),⁷¹ the Philippines (nutrition),⁷² South Africa (micronutrients),⁷³ and Zanzibar (iron supplementation and anthelmintic treatment),⁷⁴ many of which were part of multicomponent interventions and focused on children younger than 5 years rather than exclusively on the next 1000 days. On average, the reviewed nutritional interventions improved child outcomes by 0·16 SD (0·00 to 0·32; figure 1), with similarly small positive effects on health (0·14, –0·25 to 0·52; appendix p 5) and cognition (0·21, 0·07 to 0·34; appendix p 5).

Our review found that multicomponent interventions typically combine nutritional (64%), cash transfer (21%), and ECCE (50%) interventions (or a combination), and frequently also include a parenting component (50%). As shown in figure 1, these integrated programmes improve child outcomes by 0·16 SD on average (0·05 to 0·27). Effects were largest for cognition (0·16, 0·07 to 0·25) and health (0·13, –0·05 to 0·31), and smaller for social-emotional (0·07, –0·02 to 0·15) outcomes (appendix pp 5–7). Programmes combining ECCE with nutritional components have, on average, a higher effect than other multicomponent programmes (0·24, 0·00 to 0·48; appendix p 7), but still appear to be less effective than standalone ECCE and parenting interventions. These findings stand in contrast to USA-based evidence, which suggests large short-term and long-term benefits of integrated ECCE programmes such as the Perry and the Abecedarian programmes, and could reflect the difficulty of implementing complex multicomponent interventions with fidelity in low-resource contexts.^{75–77} Indeed, the lower-than-average effect found here suggests that effective integration is complex. This assertion aligns with the evidence emphasising the importance of context, risks, and population needs,^{76,78–81} as well as the quality of the components being integrated and the implementation challenges associated with the integration.

Understanding long-term effects of ECCE interventions in the next 1000 days

Given the importance of early life development for longer-term outcomes^{82,83} and the increasing global coverage of ECCE, a growing body of literature has also looked at longer-term effects of these interventions.

New publicly available datasets increasingly also allow an assessment of ECCE effectiveness across countries. Even though these datasets are cross-sectional, and thus do not identify causal effects of ECCE attendance, they allow important insights into average improvements seen among children enrolled in at-scale ECCE programmes for a diverse set of countries. One of the largest multi-country assessments is PISA, conducted in 79 mostly high-income and middle-income countries since 2000 and measuring 15-year-olds' reading, mathematics, and science abilities.⁸⁴ Analysis of the pooled PISA data suggests that conditional on country fixed effects, parental education, and parental socioeconomic status, attending 1 year of ECCE is associated with 0.15 SD higher test scores. Consistent with the existing literature on programme length and exposure,^{50,85,86} we found larger effect sizes (0.24 SD) for children attending 2 years of ECCE or more (figure 3). These effect sizes are substantial, and account for approximately 40% of the current average differences in test scores between children growing up in HICs and children growing up in upper-middle-income countries.

The effectiveness of ECCE in low-income settings can also be assessed using data from the Young Lives study, which has followed cohorts of children growing up in low-income settings in Ethiopia, India, Peru, and Viet Nam since 2002.⁸⁷ Young Lives follows two cohorts of children initially at the ages of 1 year and 8 years, collecting detailed data on ECCE and school participation, academic performance, and health behaviours. As shown in figure 4, the adjusted associations between any ECCE participation and test scores at age 15 years are similar to those seen in the PISA data, with an average difference of 0.13 SD for mathematics scores and of 0.21 SD for verbal scores for children with at least 1 year of ECCE, compared to children without any exposure. Figure 4 also shows that ECCE participation is associated with a 6% reduction (−0.057, −0.14 to 0.02) in early alcohol consumption in the Young Lives data.

The increasingly differentiated and complex set of skills that develop in the next 1000 days naturally raises the question of how much attention should be given to supporting specific areas of development within ECCE programmes. Longitudinal evidence regarding the long-term benefits of specific early life skills remains limited. Our global literature search was able to identify only 18 longitudinal studies linking measures of childhood development in children younger than 5 years to adolescent outcomes, all of which use data from high income countries (HICs; appendix pp 17–19). Virtually all studies reported positive and significant associations between early childhood development and adolescent academic performance or test scores. As figure 5 shows, early life cognitive skills were most predictive of adolescent academic performance, with an average increase of 0.23 SD (0.14–0.32) in test scores for a 1 SD increase in early life cognition. Positive but smaller

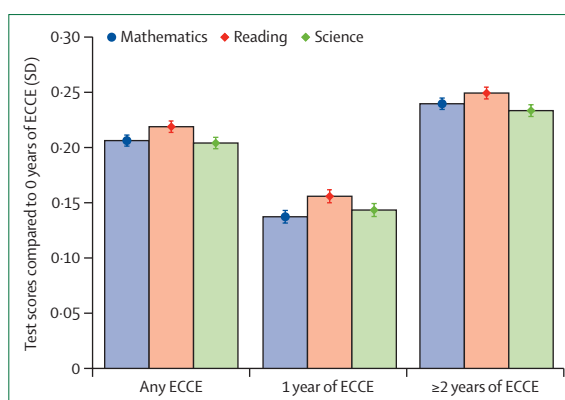


Figure 3: Associations between ECCE enrolment and academic performance at age 15 years

Results are based on ordinary-least-squares regressions, using data from 79 countries that participated in at least one of the four rounds of PISA (2012, 2015, 2018, and PISA for Development). All models compare children participating in ECCE programmes to children not participating in ECCE and control for child age and sex, household wealth index, and maternal and paternal educational attainment. All models also include country and PISA-round fixed effects. Displayed coefficients correspond to standard deviations in the outcome variables, and error bars indicating 95% confidence intervals. ECCE=early childhood care and education. PISA=Programme for International Student Assessment.

associations with adolescent academic performance were found for early social-emotional skills (0.15, 0.10–0.21) and early executive-functioning skills (0.06, 0.02–0.11).

These estimates should not be interpreted as evidence for prioritising cognitive over social-emotional or executive-function skills in early programming. Indeed, experimental evidence suggests that the long-term benefits of ECCE interventions on adolescent and adult outcomes (eg, adolescent risky behaviour, crime, welfare dependency, educational attainment, and labour outcomes) could be driven more by programme effects on social-emotional skills such as motivation and self-regulation than by improvements in cognition.^{53,88,89}

There is also increasing consensus that programmes targeting a broader set of malleable skills in early childhood are likely to yield larger returns in the long term than programmes promoting specific academic-content knowledge or constrained skills. Although the specific aspects of ECCE that matter most for children are likely to vary based on local resources and values,³⁶ high-quality ECCE programmes generally require both positive structural features, such as safe, well resourced classrooms, and well trained staff, and positive process features such as responsive, emotionally warm, and instructionally rigorous teacher–child interactions.^{24,90–92} Evidence from the USA and some LMICs suggests that ECCE programmes that use play-based, child-centred, pedagogical approaches to support a wide range of academic and foundational skills such as complex analytical thinking or social-emotional capacities are more likely to generate sustainable effects relative to those that

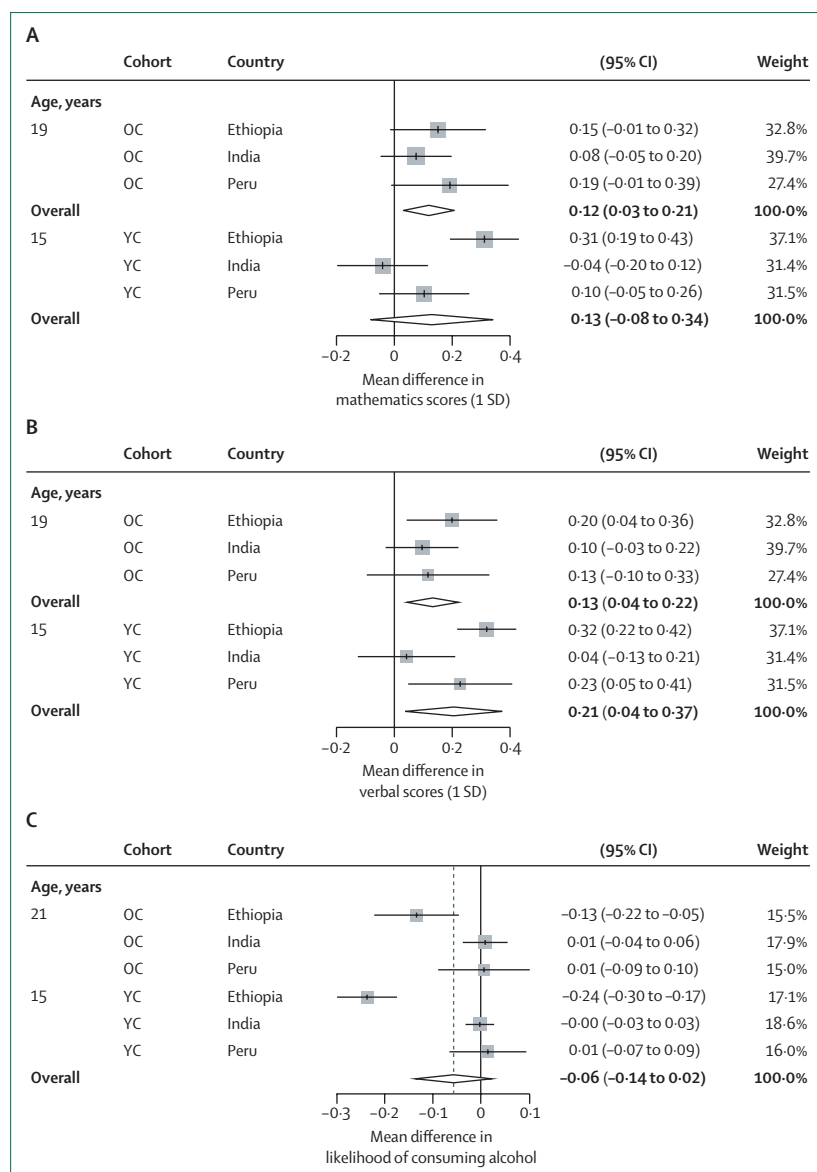


Figure 4: Estimated associations between ECCE exposure and adolescent outcomes in the Young Lives cohorts

The figure shows the mean difference in mathematics scores (A), mean difference in verbal scores (B), and likelihood of alcohol consumption (C). Children in the YC were aged on average 1 year when Young Lives was started in 2002; children in the OC were on average aged 7 years. ECCE=early childhood care and education. OC=older cohort. YC=younger cohort.

use more rote-learning methods to target narrow content knowledge such as letter identification.^{93–97} Ongoing professional development (eg, training and coaching), curricular supports, and financial supports for early childhood educators are also considered as key means for ensuring programme quality in US ECCE and some LMIC settings.^{98–101} Evidence on quality improvements and measurement of quality in LMICs is, however, limited. Evidence from LMICs within the last decade suggests that engaging parents in preschool programmes could also contribute to improved outcomes for children,^{102–104} even

though such positive effects have not been found in some settings.¹⁰⁰

Cost of interventions

Quantifying short-term and long-term benefits of programmes is crucial, but not sufficient for funders and policy makers to make sound decisions regarding programmes and investments targeting the next 1000 days. Unless funding requirements are clearly specified, the relative benefits of specific investments are hard to assess, and adequate resources to support needed programmes are hard to secure. In comparison with the rather impressive evidence base on intervention effectiveness, data on the cost of interventions in the next 1000 days are scarce, and often not directly comparable. Programme cost estimates are either based on budget and expenditure records,¹⁰⁵ or constructed based on a set of assumptions about resource needs and market prices.^{20,106,107} These approaches often result in substantially different estimates, and further accentuate the large heterogeneity in the per-child cost due to differences in quality, scale, and context.

Studies covering the provision of ECCE, cash transfers, school feeding, deworming, vitamin A supplementation, and disability screening services for children aged 3–5 years in LMICs were screened for cost data on service delivery (not considering user costs, such as fees, uniforms, transportation costs, or time spent in transit to access services). Published data on costs were extremely limited and details on costing methods were sparse. For ECCE (ie, pre-primary), average cost per child reported in the literature varied between \$21 and \$2945, with a median cost of \$289 per child and year. This large cost range is attributable to wide variations in adult-child ratios, programme structure, staff salaries, programme durations, dosages, and services, and delivery contexts. For unconditional cash transfers, the limited available data show that the transfer amounts vary largely, but suggest estimated annual costs per child that range between \$16 and \$259, with a median of \$210. The median cost for school-based or centre-based feeding programmes in LMICs was \$32 per child-year, with a wide range (\$11–288) due to the large heterogeneity in the types and volume of food provided and the modality of provision. More costing data are available for deworming and vitamin A supplementation, with a median cost of \$0.39 for deworming and \$1.75 for vitamin A supplementation (appendix p 21). Cost estimates for these two interventions show less variation than those for other interventions, even though there is heterogeneity in labour costs, numbers of doses, and distribution modalities. As for other potential interventions in the next 1000 days, such as disability screening and child security and safety, cost data are scarce, and largely originate from HICs. Importantly, much of the available cost data are for single services only rather than integrated programmes. With shared costs, such as for design,

overheads, or infrastructure for example, the costs per intervention type could, potentially, be substantially reduced.

The same holds for child security and safety, with only limited cost data mainly originating from HICs. For example, data on the costs of prevention of violence against children are complex due to variability in setting and scale, and the types of interventions, which can span from mass media campaigns to universal parenting programmes and cognitive behavioural therapy.¹⁰⁸ The available cost estimates (for 117 interventions) are summarised in the appendix (pp 21–26).

To generate cost estimates for universal ECCE programmes in all LMICs as a minimum policy target, we built on previous simulation models^{106,109} and computed child-specific, cohort-specific, and country-specific costs based on the cost estimates reported in the literature (appendix, pp 20–25). We assumed a full-day programme and a pupil–teacher ratio of 30:2 with one qualified teacher and one assistant per class. These assumptions were made with a kindergarten-like setting and group size and minimal investment framing in mind, primarily targeting children older than 3 years—smaller pupil–teacher ratios and larger investments would be needed for younger children. The table summarises the estimated per-pupil and country-level recurrent funding requirements for 1, 2, and 3 years of ECCE, with an average weighted cost of \$337 per child per year for LMICs, and \$552 globally. Assuming that a single funder (such as a central government) would fully pay for this programme, we estimated a necessary total investment for 1 year of universal ECCE ranging from 0·02 to 2·71% of gross domestic product (GDP) across all LMICs, with a weighted mean of 0·13%. This figure is in line with current Organisation for Economic Co-operation and Development expenditure levels for pre-primary education, which ranges from 0·1 to 1·1% GDP.¹¹⁰ The comparatively higher funding requirements for LMICs are due to relatively high teacher salaries (relative to GDP per capita) and much larger cohort sizes (proportion of total population ages 3–5 years) in low-income settings.¹¹⁰ Further details are provided in the appendix (pp 20–25).

Based on these simulations, the total recurrent cost of 1 year of universal ECCE coverage for all LMICs is approximately \$36·3 billion. These cost estimates do not account for children with special needs or particularly challenging contexts due to, for example, environmental, health, or political crises, but are similar to previously published LMIC estimates.¹⁰⁷

Cost of inaction

Building on the empirical literature on costs and benefits discussed previously, we turn to the economic rationale for funding these programmes from a societal or governmental perspective. The cost of inaction is the net benefits foregone in societies for not investing in

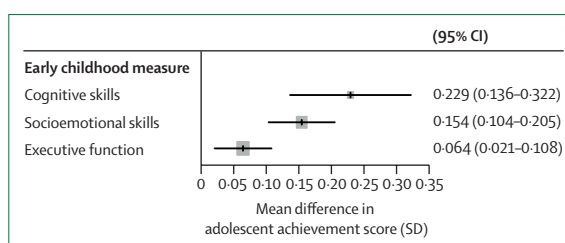


Figure 5: Estimated associations between early (aged <5 years) cognitive, social-emotional, and executive functioning skills and adolescent academic performance

Figure shows mean standardised differences in adolescent achievement score per SD increase in early childhood measure. Point estimates are based on random effect meta-analysis based on 20 cognitive, 9 socioemotional, and 23 executive functioning point estimates. Full study details are provided in the appendix (pp 17–19).

	Cost per child per year (US\$)	Total cost for 1 year (age 5 years)	Total cost for 2 years (ages 4–5 years)	Total cost for 3 years (ages 3–5 years)
Global average	522	0.08%	0.16%	0.25%
LMIC average	337	0.13%	0.25%	0.38%
Averages by income group				
Low income	97	0.42%	0.84%	1.24%
Lower-middle income	184	0.17%	0.34%	0.51%
Upper-middle income	719	0.11%	0.22%	0.33%
High income	2267	0.05%	0.11%	0.16%
Regional averages (all countries)				
Africa	181	0.30%	0.60%	0.89%
Americas	1214	0.06%	0.13%	0.19%
Asia	441	0.10%	0.20%	0.30%
Europe	1549	0.06%	0.11%	0.17%
Oceania	1453	0.06%	0.12%	0.17%
Regional averages (LMICs only)				
Africa	181	0.30%	0.60%	0.89%
Americas	294	0.08%	0.16%	0.24%
Asia	393	0.13%	0.26%	0.39%
Europe	137	0.05%	0.10%	0.15%
Oceania	234	0.47%	0.93%	1.39%

Percentages indicate % of GDP. Unit costs are weighted by population size. Wage data for 2019 were obtained from the International Labour Organization, population data from the UN, and GDP data for 2020 from the World Bank. Calculations were based on the following assumptions: pupil–teacher ratio of 30:2, wages set to 60% (teacher) and 40% (assistant) of the mean earnings of all female employees in the educational sector, a caregiver or teacher's wages to total-cost ratio of 55%, and cost per child being the same at ages 3, 4, and 5 years. ECCE=early childhood care and education. GDP=gross domestic product. LMIC=low-income and middle-income country.

Table: Summary of estimated cost and funding requirements for ECCE

recommended interventions. We focus on universal provision of at least one year of ECCE—this corresponds to SDG target 4.2 and should thus be considered the minimum global target for the next 1000 days. Specifically, we compute the cost and benefits with offering at least one year of high-quality ECCE programming to all preschool-aged children. Although the optimal timing of this single year is not obvious from a developmental perspective, placing it at the end of the 2–5-year age range would be most pragmatic¹¹¹ (taking

For more on **wage data** see ilostat.ilo.org

For more on **population data** see population.un.org

For more on **GDP data** see data.worldbank.org

into account the increasing independence of children) and would prepare them for their subsequent transition into the primary schooling system.

To monetise the benefits associated with ECCE programmes, we estimated the long-term effect of these programmes in terms of additional future wage and labour incomes likely resulting from early life participation in these programmes. As summarised earlier, the literature on the long-term effects of ECCE programmes are substantial. Apart from the Perry Preschool Program, however, few studies have been able to follow-up children into adulthood and directly quantify later wage benefits. To create longer-term links, we combined pooled estimates of ECCE effects on child development in early life from our meta-analysis with evidence on how these improvements in childhood skills relate to indicators of adult productivity. Effect estimates using either randomised or quasi-experimental study designs (from our earlier systematic review, including large scale national programmes) suggest an average increase of 0.34 SD in childhood cognitive skills due to ECCE. Following previous work,¹¹² we assumed that each SD increase in child test scores increases adult earnings by 13.1%. Based on the minimum working age in most countries, we assumed that children enter the labour market either at age 15 years or after completing the average number of years of schooling in the country, and work for 45 years. Data on the average number of completed years of schooling by countries' populations aged 25 years and older were obtained from UNESCO. Average wages were approximated as 80% of GDP per capita. Future GDP growth was based on the latest International Monetary Fund projections up to 2025;¹¹³ wages in all countries were assumed to grow by 1% per year after that, and all future incomes were discounted at 3%. Further details on the calculations are provided in the appendix (p 28). Although these estimates are for only 1 year of ECCE, our analysis of

existing programmes presented earlier, as well as most of the literature on ECCE timing, suggest substantial positive effects of a second year of ECCE.^{50,84,86,114,115}

Figure 6 presents the distribution of the cost of inaction as a share of GDP from not reaching one year of universal pre-primary coverage specified in SDG target 4.2. These figures vary substantially by country-income groups. The estimated cost of inaction is lowest in HICs (0.7% of GDP), 2.4% in upper-middle-income countries, 4.1% in lower-middle-income countries, and 6.8% in low-income countries, yielding a global average of 3.0%. The primary driver of a higher cost of inaction in lower-income countries, as well as of the relative higher cost of implementing ECCE in these settings, is the much lower baseline participation rates in these contexts (figure 2), which makes universal ECCE coverage more costly, but also results in a much larger number of children benefiting from these programmes. Children aged 3–5 years represent larger percentages of the populations in low-income countries (18.3%) compared to the other income groups (14.3% in middle-low-income countries, 10.5% in middle-high-income countries, and 7% in HICs). Lastly, the average grades of schooling attainment in the population aged older than 25 years are lower for low-income countries (4.1) versus lower-middle-income countries (6.9), upper-middle-income countries (10.0), and HICs (11.9), which implies that benefits accrue slightly earlier in low-income settings. The benefit-to-cost ratios implied in the main cost of inaction estimates are 8.7 for low-income countries, 16.8 for lower-middle-income countries, 19.1 for upper-middle-income countries, and 22.7 for HICs.

Our cost of inaction estimates are likely conservative because we did not include other child and maternal benefits of ECCE that are hard to monetise (ie, improved physical and mental health, maternal time freed from childcare, or crime reduction)^{116,117} and because we only considered private returns, omitting possible externalities to other individuals or productive factors in the economy and possible fiscal effects. Our model also only considered the direct costs incurred at the time of the intervention, although there are indirect costs (transportation or opportunity costs for parents) or costs that might be incurred later (if the intervention leads to greater subsequent schooling). We also did not include market-wide or economy-wide effects, which could attenuate or magnify the effects, depending on how aggregate labour supplies and demand are affected by the intervention. Finally, the net benefits of attending ECCE were simulated considering that the alternative is not to participate in any other intervention that might affect children's cognitive skills.

Limitations

This Series paper provides a review of global studies across interventions for which there is most evidence and is therefore not comprehensive of all aspects that relate to

For more on years of completed schooling per country see <https://data.uis.unesco.org>

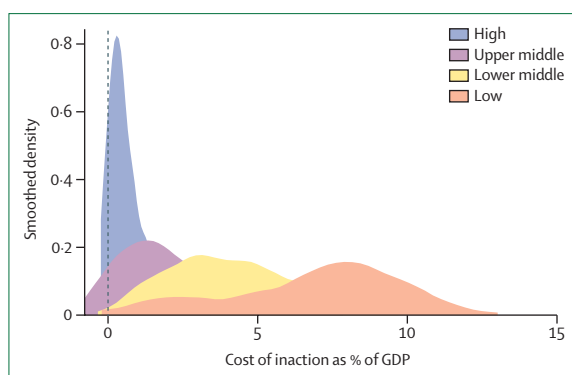


Figure 6: Distribution of the estimated country-level cost of inaction of not achieving universal coverage in SDG 4.2 as a share of GDP by country income groups

Figure shows kernel densities of the computed country-level costs of inaction by country-income groups. A density plot is a representation of the distribution of a numeric variable. It is a smoothed version of the histogram and is used in the same concept. Higher surfaces represent a larger number of countries in the respective range of the x-axis. GDP=gross domestic product. SDG=Sustainable Development Goals.

children's development in the next 1000 days. Research from LMICs continues to grow and informs our conclusions. In addition, the analyses are limited by the currently available evidence on gender, race, ethnicity, and other important characteristics that relate to children's opportunities in these years. We also acknowledge that data regarding the long-term returns to investment in the next 1000 days are scarce in LMICs, requiring us to rely on evidence from other settings to quantify future benefits in educational attainment. Similarly, published data on costs are extremely limited, and the calculations are therefore constrained to single interventions for which some data are currently available. As for cost of inaction estimates, given their forward-looking nature, these are based on several assumptions that are not easy to verify, and have therefore been investigated through a range of sensitivity analyses. Most importantly, we just focus on one year of ECCE for children aged 4–5 years. More comprehensive programmes would likely require larger investments and could thus yield different relative returns.

Conclusions for the Series and implications for policy and practice

This Series has highlighted the importance of the next 1000 days for children's development. Providing nurturing care during this period requires coordinated efforts across health, nutrition, responsive care, safety and security, and early learning. Building on previous *Lancet* early childhood development Series^{8,118,119} and adopting a socioecological life-course perspective,¹²⁰ this Series has expanded on several key messages for early childhood development into the next 1000 days.

First, inequities during the next 1000 days are persistent and likely widening as children transition into the pre-primary period. This observation is evident in the low percentage (29·9%) of children in LMICs receiving minimally adequate nurturing care during the next 1000 days. Inequities at the global level can also be seen in the small share (5%) of research on interventions coming from LMICs, as highlighted in the first paper in this Series. Global action to address these gaps is overdue, and the cost of inaction is large. Based on the cost estimates presented in this Series paper, providing access to high-quality ECCE to all children for at least 1 year would cost less than 0·15% of LMICs' GDP on average. The societal cost of not providing this, however, is large, with estimated foregone future incomes amounting to 2–7% of GDP. This figure translates to a foregone benefit-to-cost ratio of 9:1 in low-income countries, with even higher rates of return in middle-income settings.

Second, supporting children's development in the next 1000 days implies building from the predominant health, nutrition, and parenting programmes in the first 1000 days to a broader range of developmental areas with increasing complexity and heterogeneity that require the engagement of education as well as other sectors. As children's autonomy, regulatory skills, and other

socioemotional capacities further develop, and their engagement with the social and physical environment expands, there are additional behavioural facets to consider. The first paper in this Series highlighted the complexity of development in the next 1000 days and the importance of all components of the NCF to ensure healthy development over the life-course. Although the intersection of various developmental domains and behaviours creates complex landscapes for promoting health and development, these complexities make the next 1000 days a fascinating and exciting life stage that offers new opportunities for supporting young children.

Third, although the health sector has been recommended as a key entry point for interventions in the first 1000 days, ECCE settings might be more opportune for this developmental period, especially after age 3 years. As shown in the first paper in this Series,⁴ approximately 80% of interventions to promote healthy development are taking place in ECCE settings. Home-based and centre-based ECCE programmes offer platforms to promote healthy development and leverage partnerships across sectors, recognising the complementarities between children's health, nutrition, physical activity, screen time, and sleep behaviours and ECCE enrolment.¹²¹ Although centre-based programmes appear generally more attractive and feasible in the 3–5-year age range, parenting programmes can be used to support younger children and potentially be complemented^{122,123} by relevant health, nutrition, and targeted cash-transfer programmes to support children aged 2–3 years in their transition into ECCE, as well as their subsequent transition into preschool and primary school. Indeed, encouraging parent involvement in ECCE environments can ensure continuity in children's experiences across time and contexts.^{124–127} ECCE programmes might also benefit from a close collaboration with the health sector, which can be directly involved in the promotion of healthy behaviours and preventive health services within ECCE programmes. Evidence from Zambia¹²⁸ suggests that integrating health workers into primary-school programmes can yield major improvements in health behaviour and health outcomes—similar positive effects could be feasible in ECCE settings.^{129,130} Given the extensive periods of time children spend in some ECCE programmes, these programmes also offer an ideal setting to introduce children to healthy movement and eating behaviours, promoting health and catch-up growth,^{64,66,131,132} and reducing the risks of obesity and related chronic diseases later in life.

The panel summarises key recommendations for policy makers, programme implementers, and researchers in relation to children in the next 1000 days. At the policy level, a clear recognition of the crucial developmental importance of this period is needed; multisectoral programmes to support children in the next 1000 days need to be developed and promoted globally, building upon and continuing the NCF efforts in the first 1000 days. ECCE can function as a central platform in these efforts,

and adequate funding needs to be secured for supporting ECCE-based programmes at the national level. At the programmatic level, providing equitable access to adequate nurturing care that includes at least 1 year of high-quality ECCE appears to be a first feasible and potentially high-return target for most countries. These programmes can then integrate universal developmental screening, monitoring, and targeted support for vulnerable children, and serve as platforms to promote healthy behaviours, nutrition, and parent–child interactions at ECCE centres and in children's homes. In our cost of inaction analysis we focused on a single year of ECCE for all children as a first and SDG-based target. Providing optimal care for this

age group requires policies well beyond this immediate goal, potentially including home-visiting and small-group parenting programmes for younger children, cash transfers for vulnerable children, and a range of other policies to ensure safe and nurturing environments. Without these investments, achieving the SDGs for education, equity, and health outcomes appears unlikely.

Further evidence on costing integrated interventions would strengthen these efforts. Integration of health components and a life-course perspective also requires strengthening vertical and horizontal multisectoral capacity and ensuring sustainability.¹³³ Recognising the importance of specific life stages and a life-course

Panel: Recommendations for the next 1000 days

Policy recommendations

- Building on the first 1000 days, increase investment in the next 1000 days, particularly in LMICs, extending the Nurturing Care Framework and incorporating a socioecological life-course perspective.
- Commit to investing at least 0.15% of GDP towards basic interventions to support children in the next 1000 days, especially in ECCE, complementing investments in the first 1000 days.
- Establish multisectoral coordination mechanisms to ensure adequate support for policies promoting early child development in the next 1000 days and to coordinate efforts with local communities, social protection services, and other existing platforms.
- Systematically assess and track inequities in nurturing care opportunities and implement policies to reduce these at national and local levels.
- Ensure children in the next 1000 days are visible in policies and strategies on disease control, climate change, and urban planning, including provision of green spaces for play.

Programme recommendations

- Provide access to adequate nurturing care, including equitable access to high quality ECCE, safe and supportive home and family environments with adequate stimulation, learning opportunities and responsive interactions, protection from physical punishment, and adequate nutrition to all children.
- Ensure that ECCE programmes are of high quality and culturally responsive, with an emphasis on adequately paid and trained teachers, adequate teacher–student ratios, child-centred play, evidence-based curricula, and warm, stimulating, and responsive classroom interactions. This effort also requires frequent monitoring and investing in efforts to ensure high quality.
- Implement universal developmental screening and monitoring and establish referral and support services for children with developmental delays, disabilities, and other vulnerabilities.
- Increase multisectoral collaboration engaging maternal, paternal, and other caregivers to provide protective environments for children, with particular attention to

family mental health, violence prevention, and the creation of safe spaces for children to play.

- Provide financial support to the most vulnerable populations, particularly through cash transfer programmes given their proven effectiveness.
- Routinely integrate costing policies into programme planning and implementation.

Research recommendations (funders and researchers)

- Promote research that acknowledges the fundamental importance and multidimensionality of development and health behaviours in the next 1000 days.
- Improve measurement of nurturing care and healthy development indicators.
- Collect nationally representative data on children's development in the next 1000 days (eg, through UNICEF Multiple Indicator Cluster Surveys), with a particular focus on vulnerable subpopulations and culture-specific and context-specific early childhood development skills.
- Collect information on programme quality, costs, and implementation features (eg, ECCE, parenting and violence prevention, disability screening, and nutritional programmes) that are crucial for replication, quality, and scalability.
- Increase research on contextually relevant interventions to promote healthy development and healthy behaviours, especially for vulnerable children and those in LMICs, with attention to variability in enablers and barriers.
- Expand research on the integration of nutritional and parenting efforts with ECCE, as well as on the impact of ECCE programmes on parent and family wellbeing.
- Examine the next 1000 days and the achievement of SDGs in relation to gender, race and ethnicity, poverty, and the intersection of these.
- Increase research on the effects of pollution and climate change on the development and health of children in the next 1000 days, and ways to mitigate these risks in already vulnerable settings (eg, very low-income and conflict and refugee settings).

ECCE=early childhood care and education. GDP=gross domestic product. LMICs=low-income and middle-income countries. SDG=Sustainable Development Goals.

approach, new programmes should try to improve the continuity and alignment between the first and next 1000 days and link with the primary school years that follow.

As for research, knowledge of what works, for whom, and in what contexts, remains incomplete in the next 1000 days. Evidence is particularly lacking for migrant and displaced populations, for programmes preventing violence and harsh punishment, and efforts addressing climate change and pollution, as well as for children with disabilities or developmental delays, who remain underserved and understudied in LMICs. Additional research on family-based interventions to promote responsive care, safety, and security, but also to promote healthy behaviours, as well as research on the ideal design of ECCE programmes from a socioecological life-course perspective is needed. Much more data is also needed regarding the cost of intervention—without routine, robust, and standardised collection of cost data identifying the most cost-effective ways to improve child outcomes will not be possible.¹³⁴ Future work is also required to better examine the next 1000 days and the achievement of SDGs in relation to gender, race and ethnicity, poverty, and the intersection of these.

The next 1000 days are a key stage of children's development and of central importance for their life-course trajectories. Protecting children against environmental and psychosocial risks during this period and providing nurturing care can help to sustain the gains made in the first 1000 days but also help further improve children's life-course trajectories. Increased attention on the next 1000 days is imperative for the achievement of the SDGs, and the time for global action is now.

Contributors

CED conceptualised and coordinated the Series with inputs from the Series steering committee members (AKY, GF, MN, ADO, and MST). MN and GF conceptualised this paper in consultation with the rest of the Series steering committee members (CED, AKY, ADO, and MST) and advisory group. MN, GF, EG-W, SO, CV, FL-B, JC, DCM, RB, and MJL did the literature reviews. MN, GF, CV, EG-W, SO, JC, MJL, FL-B, JB, and RB did the data analysis, interpretation, or both. MN, GF, CV, EG-W, SO, MJL, JC, CV, FL-B and JB contributed tables, figures, or both. All authors reviewed the drafts, made crucial additions and edits, and approved the final submission.

Declaration of interests

We declare no competing interests.

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Editorial note: The Lancet Group takes a neutral position with respect to territorial claims in published maps.

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